Practical 4 — Understanding Data with Graphs

Author: Student

Dataset: AER `TeachingRatings`

Observations: 463 evaluated courses taught by 94 professors

Question 1. Duplicate Cases and Age Analysis

Theoretical Background

When evaluating observational data collected at the course level, an individual professor who teaches multiple courses appears multiple times in the dataset. If we calculate summary metrics (such as average age) across all course records, professors with higher course loads are over-weighted.

To obtain an accurate representation of faculty demographics, we deduplicate the dataset using the instructor identifier variable (`prof`), restricting the sample to one observation per unique professor (N = 94).

Statistical Summary

Data Group	Observations	Mean Age (Years)	Std. Deviation (Years)
All Observations	463	48.3650	9.8027
Unique Instructors	94	47.5532	10.2565

Interpretation

The average age across all course observations is 48.37 years, whereas the mean age for unique professors is 47.55 years. The slightly higher mean in the full course dataset indicates that older professors in this sample taught slightly more courses on average than younger professors.

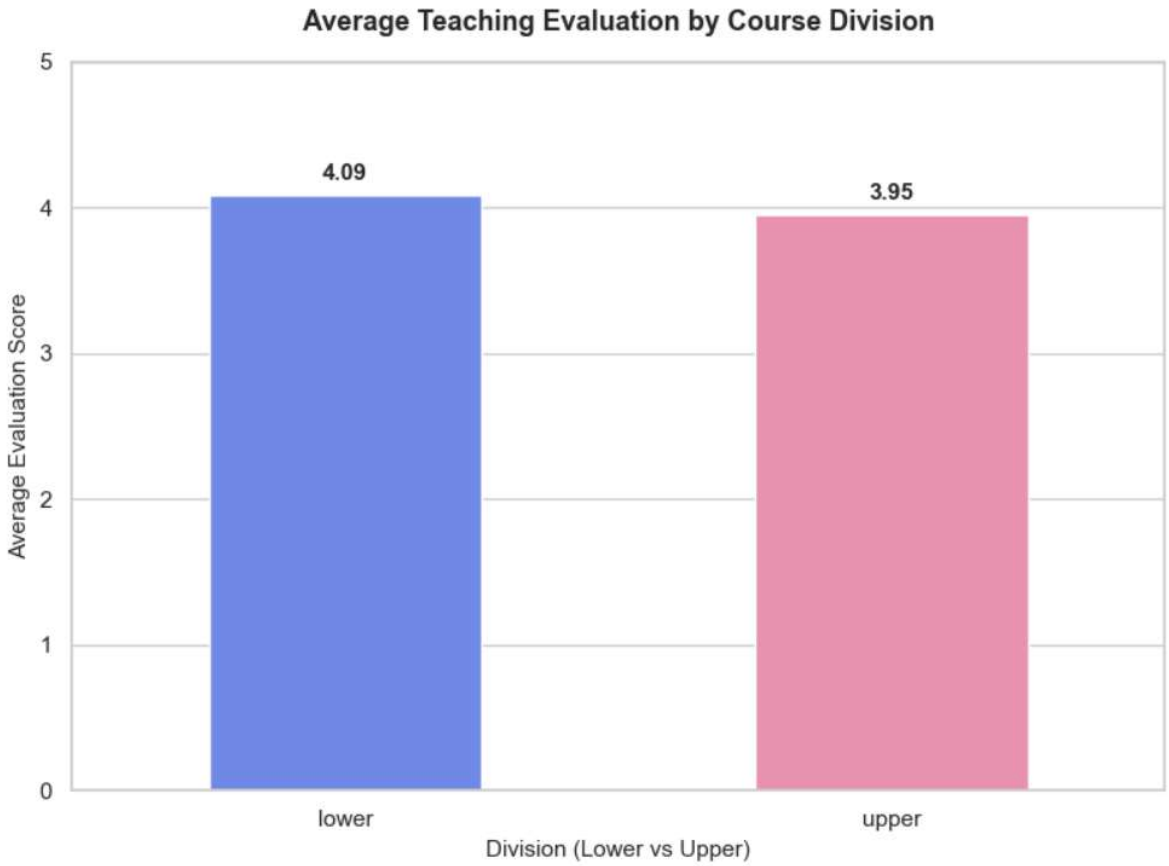
Question 2. Course Division and Teaching Evaluations

Theoretical Background

Course levels are broadly split into lower-division (introductory/general education) and upper-division (advanced/major-specific) courses. Comparing average evaluation scores across these divisions reveals whether course level influences student grading behavior.

Summary Data

Division	Average Evaluation Score
Lower	4.09
Upper	3.95



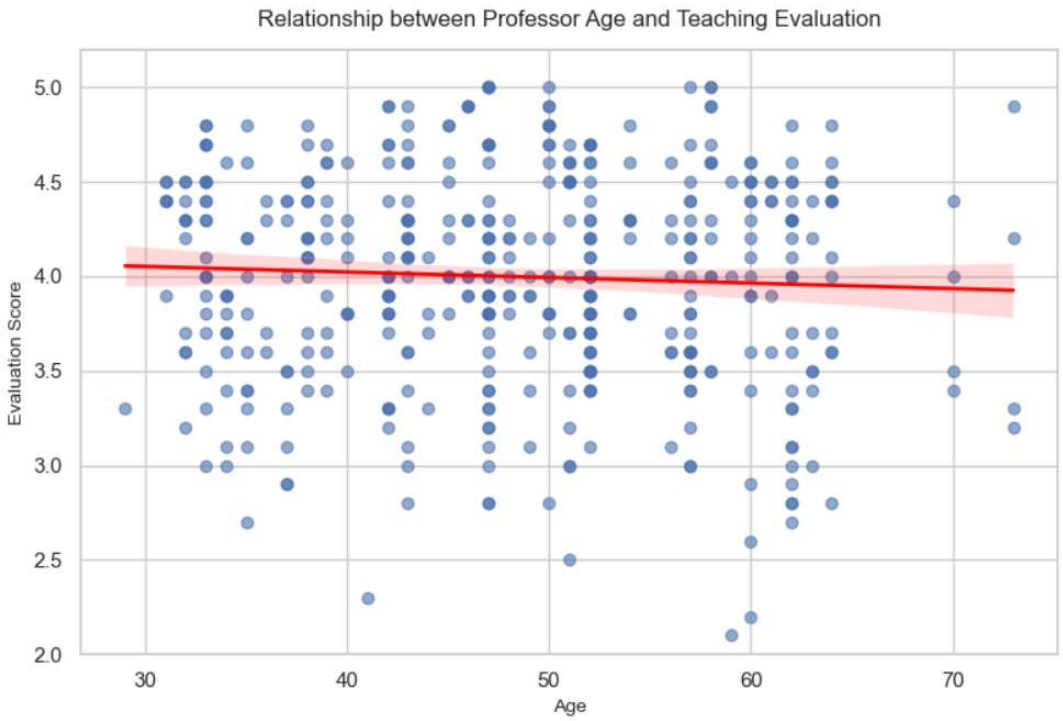
Findings

Instructors teaching lower-division courses received a slightly higher average teaching evaluation (4.09) compared to those teaching upper-division courses (3.95)

Question 3. Relationship Between Age and Teaching Evaluation

Theoretical Background

To evaluate how instructor age relates to student teaching evaluations, we construct a scatter plot paired with an Ordinary Least Squares (OLS) trendline. This allows us to assess both overall data dispersion and linear directionality



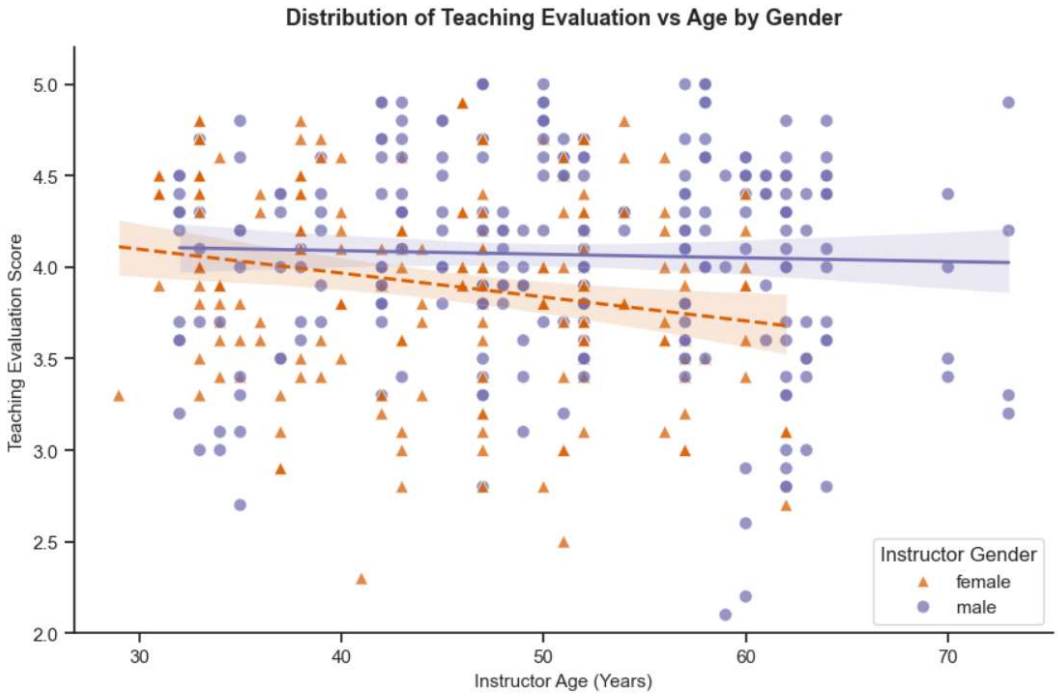
Findings

The scatter plot demonstrates widespread evaluation ratings across all age brackets. The fitted regression line shows a very modest negative slope, indicating that older professors receive slightly lower average evaluation scores in this dataset.

Question 4. Gender-Differentiated Analysis

Theoretical Background

Stratifying the bivariate relationship between age and evaluation scores by gender enables us to observe whether rating distributions and age patterns differ between male and female instructors.



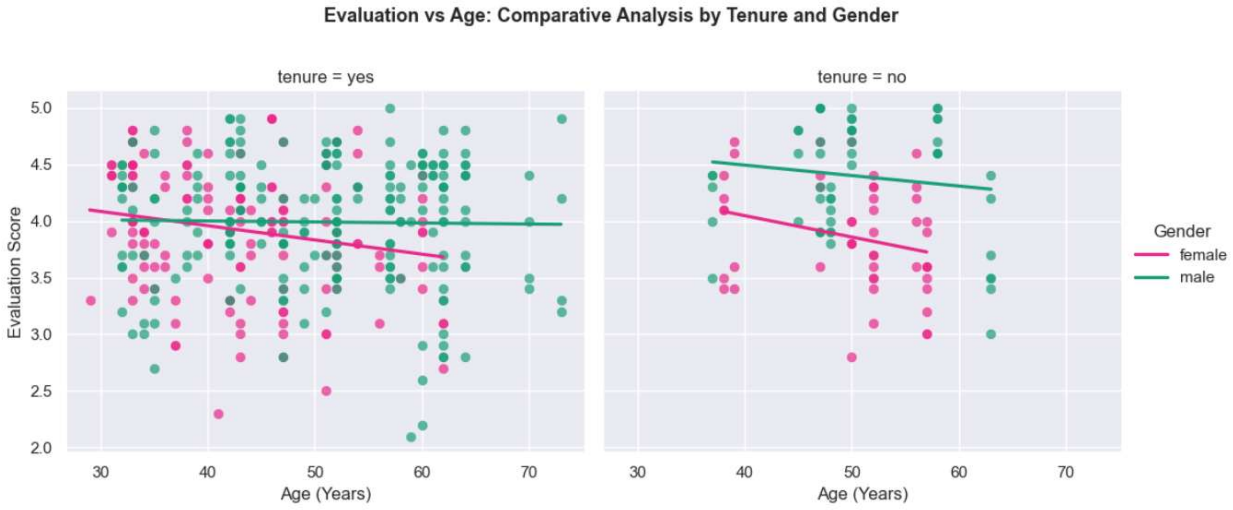
Findings

Both male and female instructors exhibit a wide distribution of scores across the entire age spectrum. However, male instructors are more heavily represented in the upper evaluation tiers (> 4.5) across multiple age groups.

Question 5. Gender and Tenure-Differentiated Analysis

Theoretical Background

To examine intersecting structural factors, we incorporate both `gender` (via marker color) and `tenure` status (via marker shape: circles for tenured, crosses for non-tenured) into a single scatter plot.



Findings

- Non-tenured faculty (`tenure = no`) cluster predominantly in younger age categories (< 45years).
- Tenured faculty (`tenure = yes`) extend into higher age categories (> 50 years).
- Female non-tenured professors tend to receive varied ratings, whereas male tenured faculty account for most high-evaluation data points in the 50+ age bracket.